SAVITRIBAI PHULE PUNE UNIVERSITY

DEEP LEARNING (410251)

BE (Computer Engineering) --- Semester VIII --- 2019 Pattern

Total Marks: 70 | Time: 2½ Hours

Instructions:

1. Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
2. Figures to the right indicate full marks.
3. Neat diagrams must be drawn wherever necessary.
4. Make suitable assumptions whenever necessary.

Unit IV --- Recurrent Neural Network --- 17 Marks

Q3) Solve **any one** of the following:

a) Explain Recursive Neural Network. How is it different from Recurrent Neural Network? Give an application of each. [6]

b) Explain Long Short-Term Memory (LSTM) in detail. Draw and explain the cell architecture with forget, input, and output gates. [6]

c) Explain in brief the working of a Recurrent Neural Network. What is the role of the hidden state? How does an RNN handle variable-length sequences? [5]

OR Q4) Solve **any one** of the following:

a) Differentiate between CNN and RNN. Compare their architectures, use cases, and training approaches. [6]

b) What are the challenges of long-term dependencies in RNNs? Explain the vanishing and exploding gradient problems. How do gated architectures (LSTM, GRU) alleviate these? [6]

c) Explain the Encoder-Decoder RNN model for sequence-to-sequence tasks. How is it used in machine translation? Discuss attention mechanism. [5]

Unit V --- Deep Generative Models --- 18 Marks

Q5) Solve **any one** of the following:

a) Explain Deep Generative Models. How do they differ from discriminative models? Discuss applications in image generation and data augmentation. [6]

b) Explain Boltzmann Machine in detail. Discuss its architecture, energy function, and training. How is a Restricted Boltzmann Machine (RBM) different? [6]

c) Explain Generative Adversarial Network (GAN) with an example. Describe the minimax objective function and the training process. [6]

OR Q6) Solve **any one** of the following:

a) Explain Deep Belief Networks (DBN) in detail. How are they constructed from stacked RBMs? Discuss the greedy layer-wise pretraining approach. [6]

b) What is a Generative Adversarial Network? Explain its generator and discriminator components. Draw the architecture showing the adversarial training loop. [6]

c) Explain types of GAN: DCGAN, Conditional GAN (cGAN), and CycleGAN. Compare their architectures and applications. [6]

Unit VI --- Reinforcement Learning --- 17 Marks

Q7) Solve **any one** of the following:

a) Explain Markov Decision Process (MDP). Define the key components: states, actions, transition probabilities, reward function, and discount factor. How does policy iteration and value iteration work? [6]

b) Explain Deep Reinforcement Learning. How does combining deep neural networks with RL address the limitations of traditional RL? Discuss the Deep Q-Network (DQN) architecture. [6]

c) What are the challenges of Reinforcement Learning? Discuss the exploration-exploitation tradeoff, delayed rewards, and sample efficiency. [5]

OR Q8) Solve **any one** of the following:

a) Explain the process of Deep Q-Learning. Describe the Q-learning update rule, experience replay, and target network. How do these stabilize training? [6]

b) Explain Reinforcement Learning for the Tic-Tac-Toe game. Model it as an MDP and describe the state representation, reward structure, and learning process. [6]

c) Explain the Dynamic Programming algorithm for Reinforcement Learning. Compare policy evaluation, policy improvement, and policy iteration. [5]

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